

Modeling interception movement distributions

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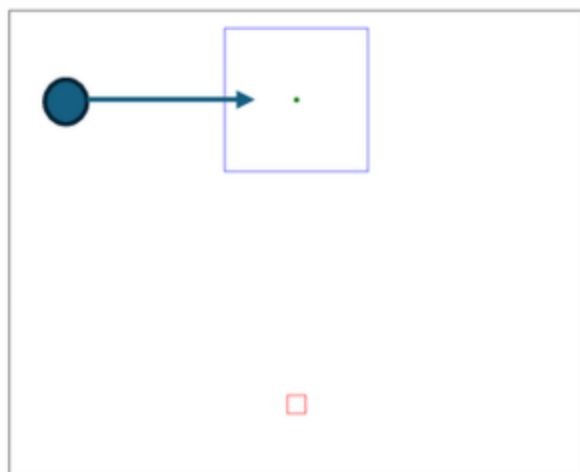
A compact personal representation for stochastic movement generation

The task and the evaluation split

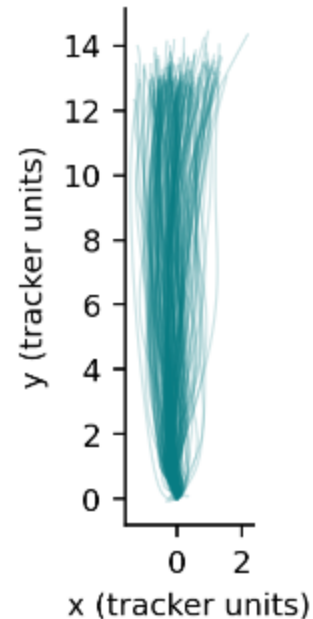
4,732 trials from 28 participants

Four rotations: 17 train / 4 validation / 7 test

Task display



subject01: 180 recorded trials



Free eye movement. All 180 trajectories from the first participant by identifier. Median retained trials: 179.5.

The compression pipelines

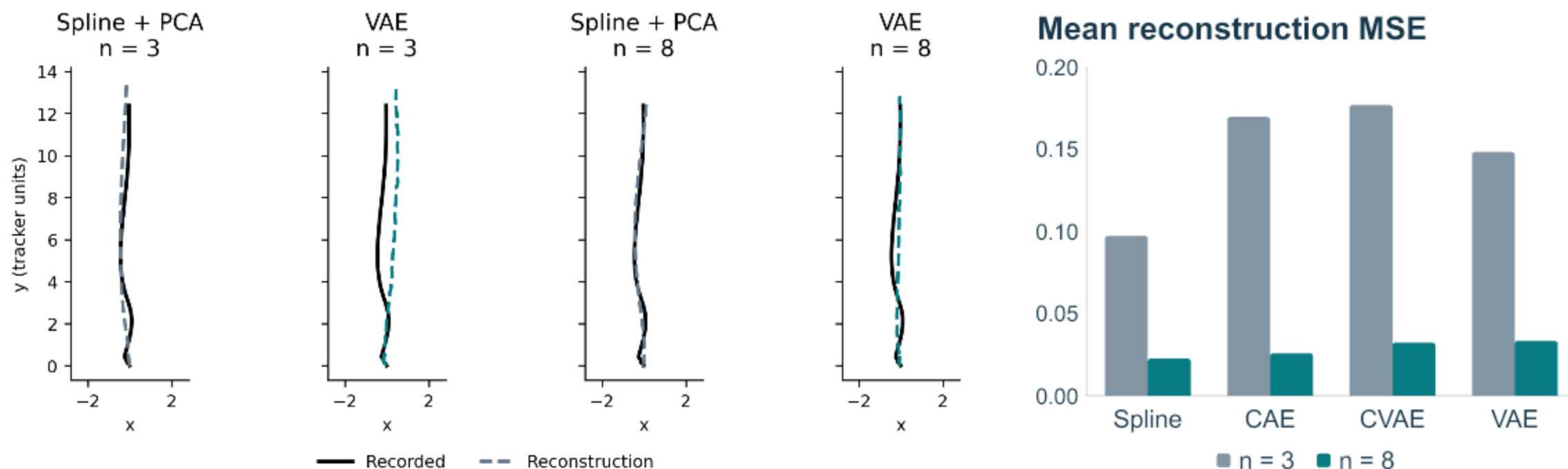
Every trajectory becomes 100 x 2 positions on normalized time.

Stage	Spline + PCA	Neural models
Input	100 x 2 positions	200 standardized coordinates
Encoder	9 coefficients per axis; PCA on N x 18 matrix	Two 256-unit layers; n-value code
Decoder	Inverse PCA; two spline basis products	Two 256-unit layers; 200 coordinates
Timing output	Ridge on code + 5 conditions	Separate two-duration head

CVAE and CAE use five task conditions. VAE masks them to zero. Matched $n = 3$ and 8.

Reconstruction favors spline + PCA

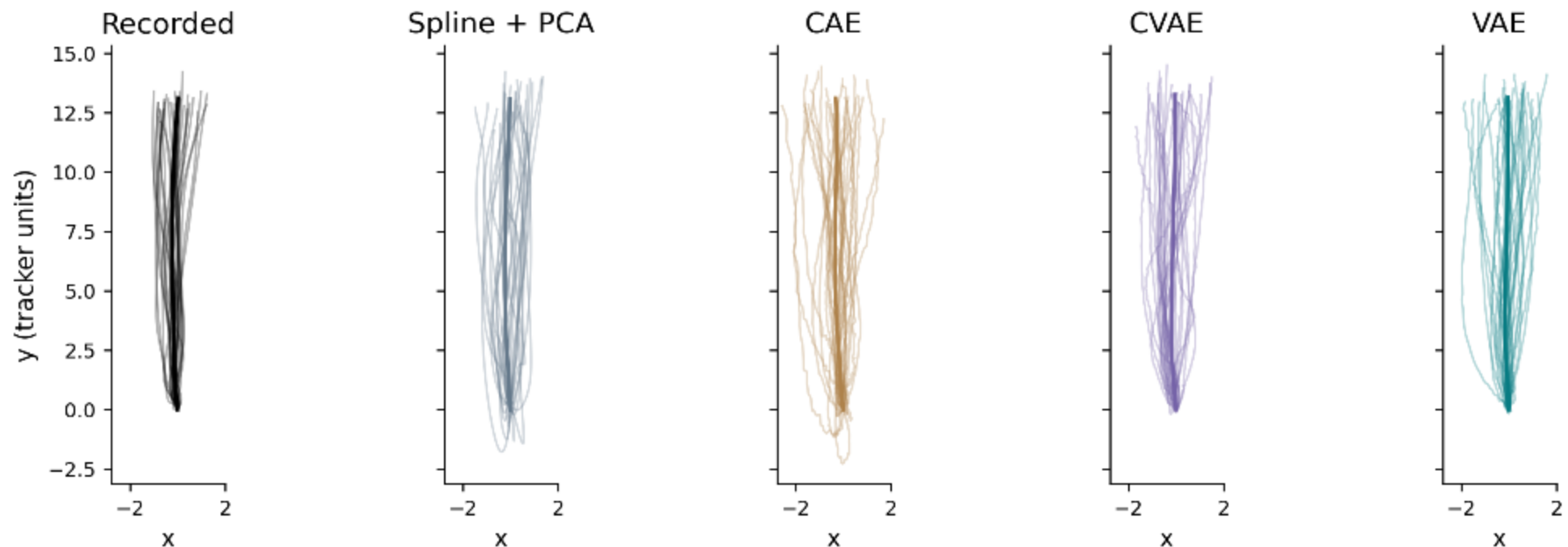
The encoder receives the observed test trajectory.



Black: recorded; dashed: reconstructed. One median-duration trial. Lower MSE is better. Spline beats each neural model under BH-140; spline vs CAE $n=8$ does not survive Holm.

Generation uses separate context trials

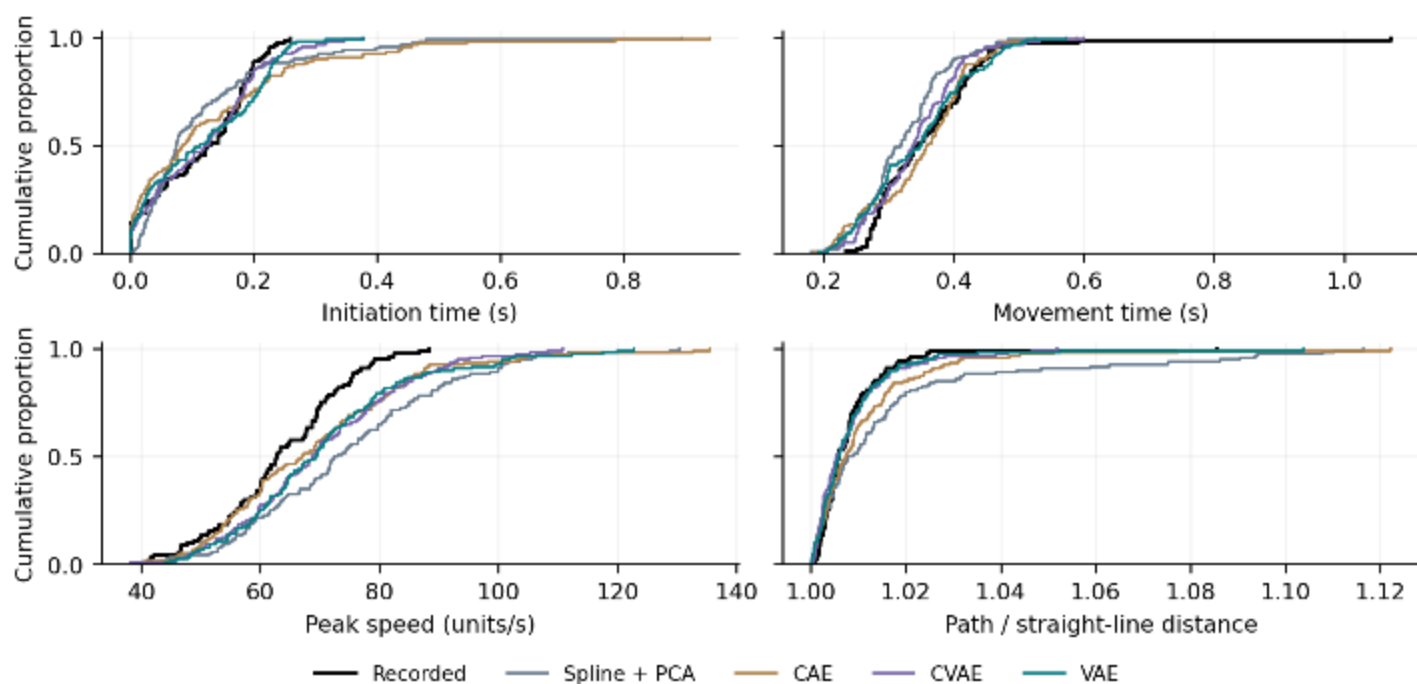
Context codes establish a personal center. We decode 120 draws around it using shared training covariance.



Same participant, separate query recordings. $n=8$, all models. Both capacities appear in the report. Thin: first 30 paths. Thick: mean of all paths. Equal x/y scale. These examples do not determine the cohort ranking.

Generated kinematic-feature distributions

Cumulative curves show individual features. Cohort KS averages all eleven feature discrepancies.



Mean feature KS

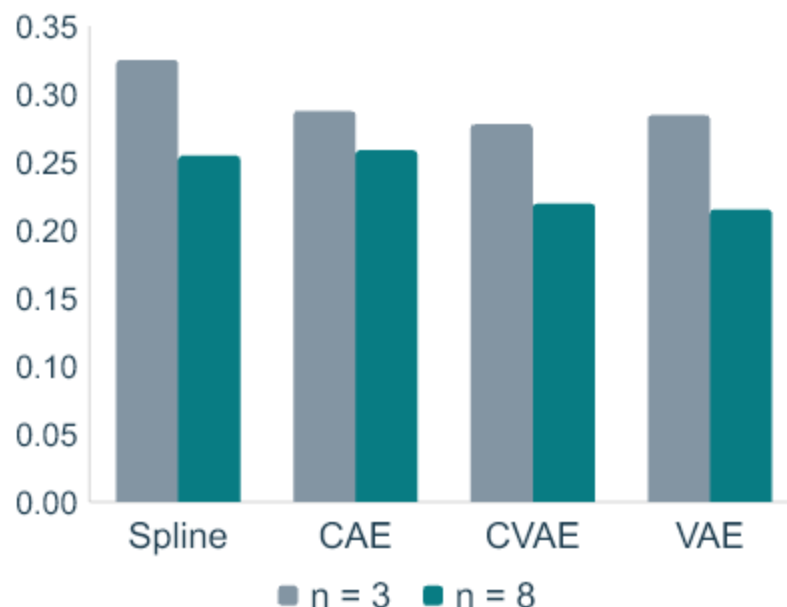
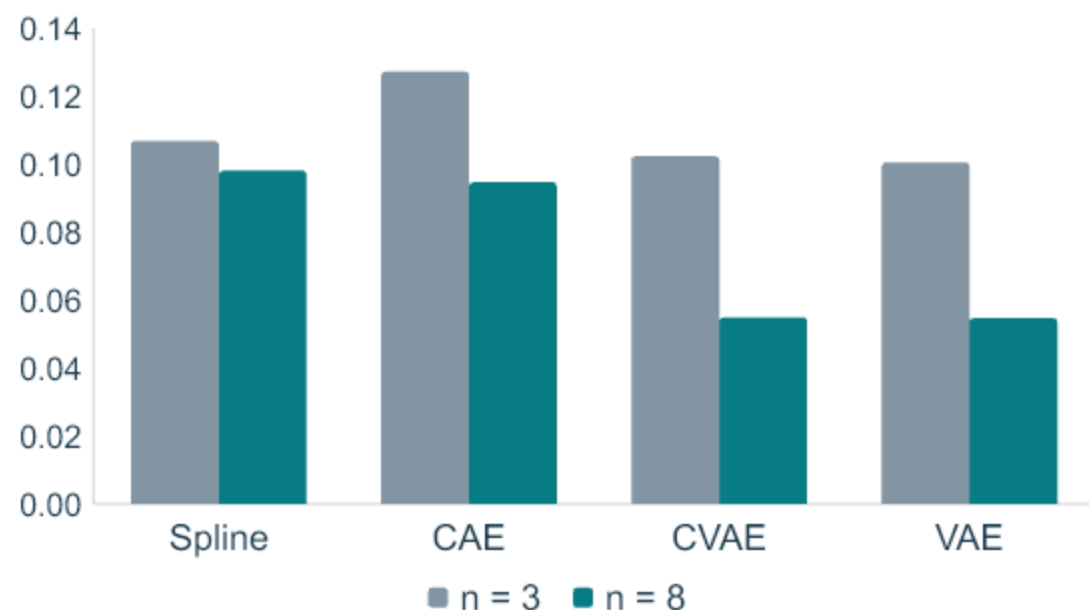


Illustration: subject01, $n=8$. VAE/CVAE beat spline on all three feature metrics at both capacities (BH and Holm-140). VAE has no uniform significant advantage over CVAE.

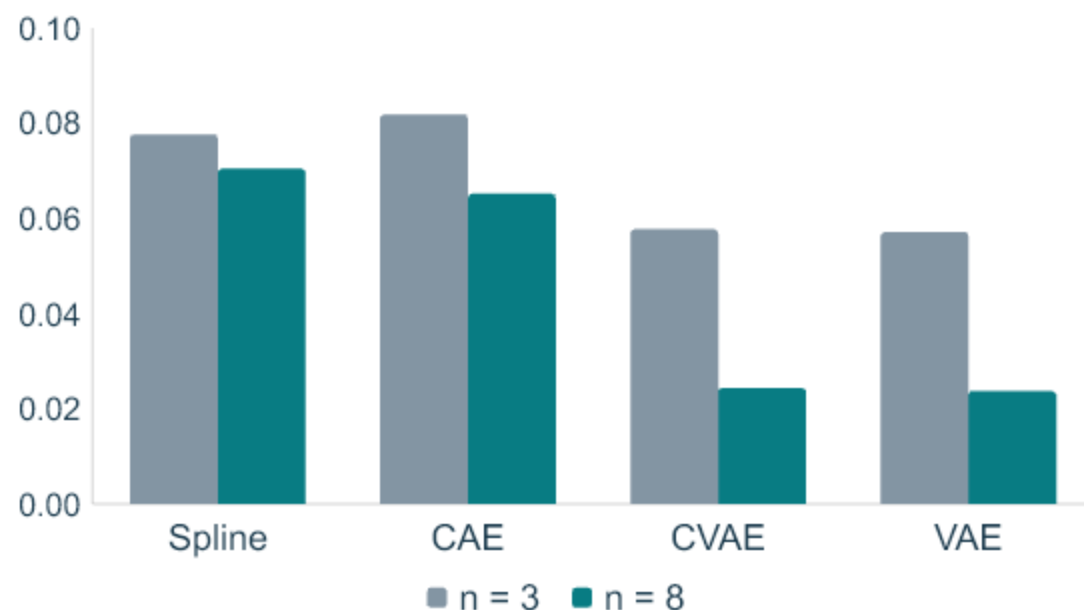
Full-trajectory generation supports VAE and CVAE

Energy and MMD compare complete sets of ordered paths, including their variability.

Full-path energy



Full-path MMD squared

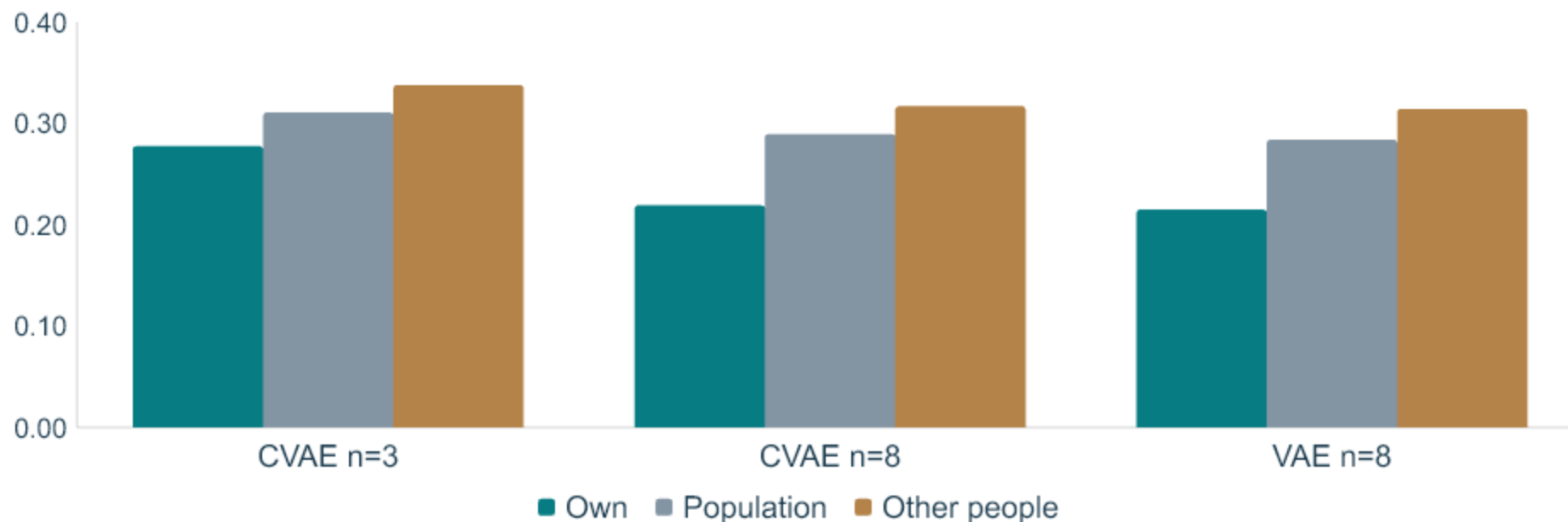


Lower is better. At $n=8$, both VAE/CVAE beat spline (BH and Holm-80), also with balanced axes. VAE vs CVAE: $q=.672$ and 1.000 . At $n=3$, results depend on metric and scaling.

The participant's own center improves generation

We change the center while holding decoder, noise and task draws fixed.

Mean feature KS after replacing the center



Mean feature KS; lower is better. All VAE control comparisons: BH $q < 1.6 \times 10^{-6}$ (6 tests). All CVAE comparisons: $q < .000382$ (12 tests).

What the compact model achieves

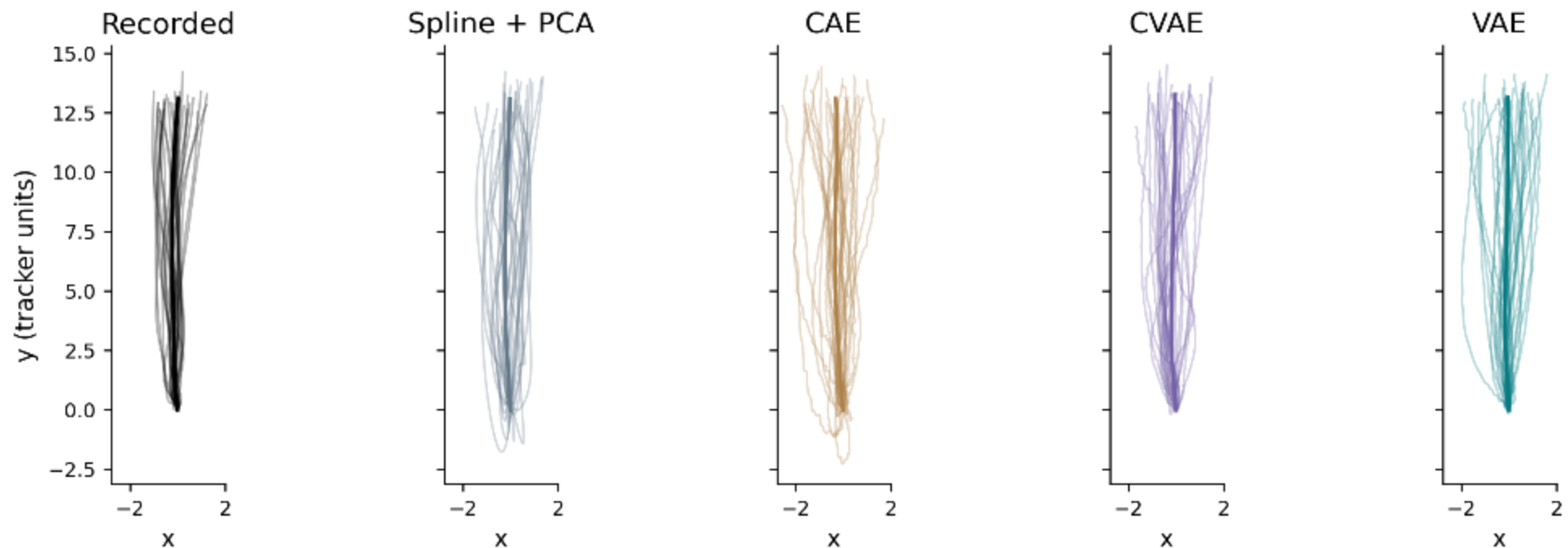
Stored context resampling has lower mean path discrepancy.
It retains complete paths, while the compact models retain a personal center.

Raw path reference	Energy	MMD ²
VAE n=8	0.0548	0.0238
Resample stored context	0.0328	0.0054
Repeat context mean	0.4621	0.2452

No formal test against these references. Reducing 11 correlated features to 9 preserves VAE/CVAE over spline, but weakens a unique VAE claim. No application acceptance threshold was established.

Dashboard demonstration

VAE $n=8$ default. Compare $n=3$, CVAE and spline + PCA. Finish with cohort benchmarks.



Recorded and generated examples if the live demo is unavailable. Same-session variability, with no claim that each latent coordinate represents a named strategy.